Yan **Li**

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Research Interests

Trustworthy Medical AI; Multimodal AI for Healthcare; Medical Vision-Language Models; Clinical AI Agents.

Education

Beijing University of Posts and Telecommunications Beijing, China

M.S. in Management Science and Engineering Sept. 2025–present

Advisor: Yawen Li; Co-advisor: Guanhua Ye.

University of Electronic Science and Technology of China Chengdu, China

Dual B.S. in Finance and Electronic Information Engineering Sept. 2021–July 2025

Thesis Advisor: Bo Liu.

Research Experience

CAPRA: Multimodal Auditing under Missing Clinical Metadata ACM MM 2026 Oral

- Led CAPRA, a semantic-visual multimodal framework that fuses medical images with proxy-axis tokens and calibrated posteriors for missing-metadata subgroup auditing.
- Validated fundus, dermoscopy, and chest-radiography tasks under distribution shift, improving worst-group accuracy in 14 of 15 learner–dataset settings.

MedFuse: Anatomical-Prior Fusion for Retinal Lesion Segmentation FCS 2026

- Led technical development with an ophthalmologist at Peking University First Hospital, integrating RGB fundus features with retinal vascular priors for diabetic-retinopathy lesion segmentation.
- Evaluated a hierarchical dual-encoder fusion pipeline on DDR and IDRiD across five backbones, including annotation-free priors from a multimodal large language model.

Teaching

Beijing University of Posts and Telecommunications

Big Data and Intelligent Management Fall 2025, Fall 2026

Teaching Assistant; co-designed a 130-student course for early-stage master's and PhD students.

Peer-Reviewed Publications

Beyond Metadata: CAPRA for Hidden Subgroup Analysis under Missing Metadata in Medical Imaging
Yawen Li, **Yan Li**, Zhe Xue, Yingxia Shao, Meiyu Liang, and Guanhua Ye*.
Proceedings of the 34th ACM International Conference on Multimedia (**ACM MM**), 2026. Paper
Oral presentation. Selected as one of **162** main-track oral papers from **7,053** reviewed submissions.

What Does Your Short-Answer VQA Score Actually Measure? Evaluator-Dependent Instability in Multimodal Short-Answer Benchmarks
Guanhua Ye, Jingbin Niu, **Yan Li**, Meiyu Liang, Zhe Xue, Yingxia Shao, and Yawen Li*.
Proceedings of the 34th ACM International Conference on Multimedia (**ACM MM**), 2026. Paper

Rethink Representation Learning for Questionnaire Data
Guanhua Ye, Jifeng He, **Yan Li**, Junping Du, Zhe Xue, Yingxia Shao, Meiyu Liang, and Yawen Li*.
Proceedings of the AAAI Conference on Artificial Intelligence (**AAAI**), 40(33):27782–27790, 2026. Paper

FeBT: A Feature Balancing Transformer for Corporate ESG Forecasting
Yawen Li, Mengyu Zhuang, Guanhua Ye*, **Yan Li**, Junheng Wang, Jinyi Zhou, and Pengfei Zhang.
IEEE Transactions on Knowledge and Data Engineering (**TKDE**), 2025. Paper
Outstanding Paper Award, 3rd Academic Conference on Data Intelligence and Management (2023).

MedFuse: A Multi-source Data Fusion Framework for Diabetic Retinopathy Lesion Segmentation
Jiantong Du, **Yan Li**, Yawen Li, Liwen Liao, Zhihao Zhao, and Guanhua Ye*.
Frontiers of Computer Science (**FCS**), 20(3):2003625, 2026. Paper

Conference Presentations

"Hidden Subgroup Analysis under Missing Metadata in Medical Imaging" Nov. 2026
Oral Presentation at ACM MM 2026, Rio de Janeiro, Brazil. Poster | Slides

"Auditing Historical Clinical Queries with Missing Availability Time" Aug. 2026
Oral Presentation at CCF BigData 2026, Harbin, China. Paper | Slides

"Slot-Based Evidence Routing for Budgeted Multi-Agent Coordination" May 2026
Oral Presentation at CINT 2026, Hangzhou, China. Paper | Slides

"Trust-free Personalized Decentralized Learning" Feb. 2026
Oral Presentation at IEEE BigComp 2026, Guangzhou, China. Paper | Slides

Honors and Awards

University of Electronic Science and Technology of China

Certificate for Honors Research 2025
"Academic Star" Award of School of Management and Economics (sole recipient) 2024
Outstanding Student Scholarship 2024
Model Student Scholarship 2023